



Fire-detecting Autonomous and Controlled Car

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Abstract

As industrial infrastructure expands rapidly across Egypt, fire safety has become a critical concern. Central Agency for Public Mobilization and Statistics (CAPMAS) reported 46,925 fire accidents in 2024, resulting in over 1,000 casualties. By 2025, this figure rose by 8.7% to approximately 51,000 accidents. This alarming trend necessitates a shift toward unconventional, technology-driven fire detection mechanisms. The objective of this project is to develop a dual-response fire detection vehicle capable of switching between

- controlled mode: utilizes Bluetooth for precise, short-range human intervention
- autonomous logic-based navigation: employs deterministic, If-Then architecture to detect fire sources without the need for external connection.

Such a hybrid approach ensures operational reliability and human safety in high-risk, high-heat emergency environments and provides a cost-effective, scalable solution that reduces response time and prioritizes human life, especially in industrial settings.

Materials

To implement the project, we used the following:



Recommendations

Due to the time frame and budget limits, the project was carried out using the aforementioned components and to only address the detection role. But, it provides insights into alternatives and role expansion based on presented performance:

- As for the connection of the Bluetooth, its range is limited with respect to certain parameters such as distance and barriers. We suggest upgrading to a WiFi-Module (e.g., ESP32/WiFi mesh) for greater range and stability.
- As for Navigation using the Ultrasonic sensor, it possesses several blind spots; for example, soft materials don't reflect sound well. We suggest using Lidar or a camera for better obstacle avoidance.
- The algorithmic If-Then logic can be further expanded to a full simultaneous localization and mapping (SLAM) program using machine learning to allow the vehicle to learn and patrol paths on its own, rather than just relying on collision avoidant random-walk navigation.

Methods

The development cycle began with a simulation in TinkerCAD and Cirkuit to validate the control logic and circuit integrity. Upon virtual testing, We moved to physical construction of the system. This involved mounting the 4-motor array to the chassis within a custom-built maquette to ensure that the work-energy requirements calculated during modeling translated successfully to real-world navigation.

- Movement and Torque

To meet the power demand of the 4 DC motors, the L298N motor driver was integrated with an Arduino Uno. This configuration utilizes Pulse Width Modulation (PWM) to regulate torque required to overcome friction. Applying the work-energy principle, the system ensures that the work done by the motors (W_{app}) exceeds the sum of frictional resistance, maintaining the velocities required for movement.

- Fire Detection

We employed fire detection mechanism, starting with input from the flame sensor to the Arduino. The flame sensor detects either infrared (IR) or ultraviolet (UV) rays emitted from fire, converting this signal into an electrical signal.

- Navigation

How the car navigates its surroundings depends on whether it is connected via Bluetooth or not. If so, it follows the orders of the controlling app as the person sends them; if not, it depends on the ultrasonic, a pre-determined route, and the if-then logic described in Fig.1.

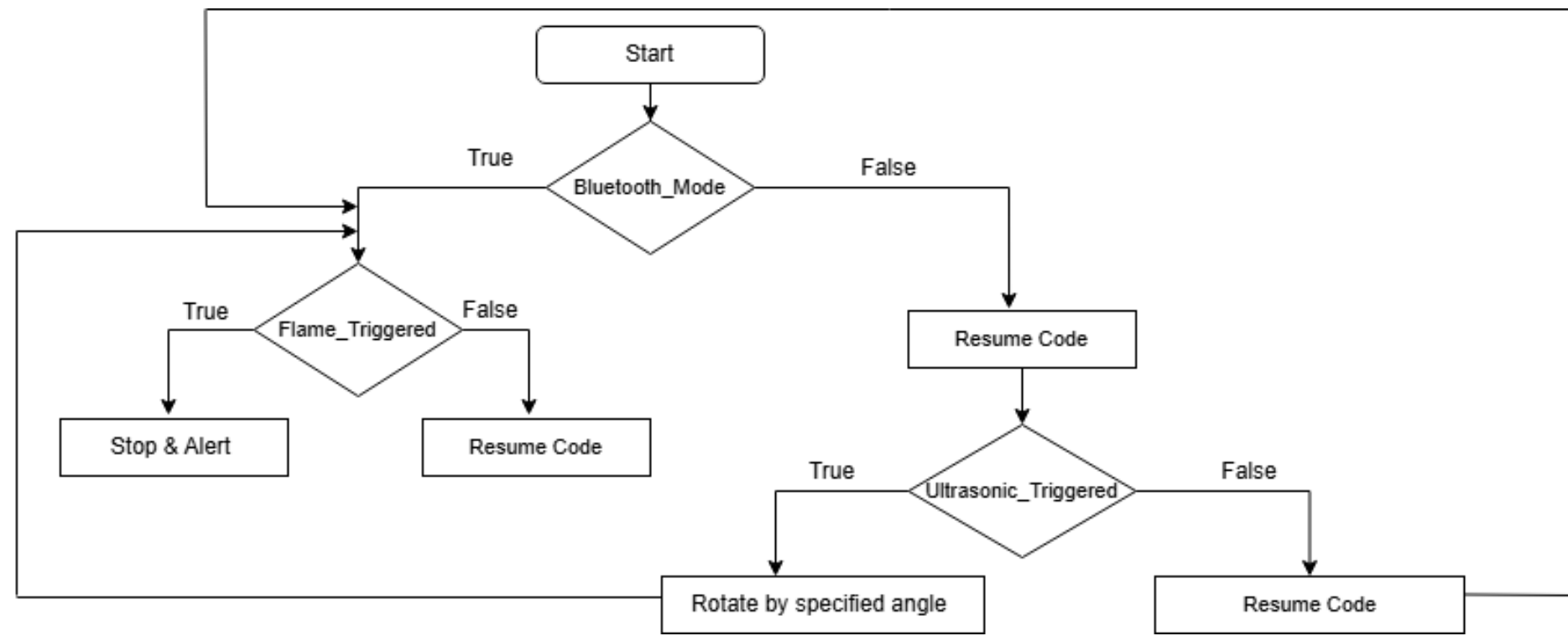


Fig 1: If-Then architecture for obstacle avoidance and fire detection. (Repeated indefinitely)

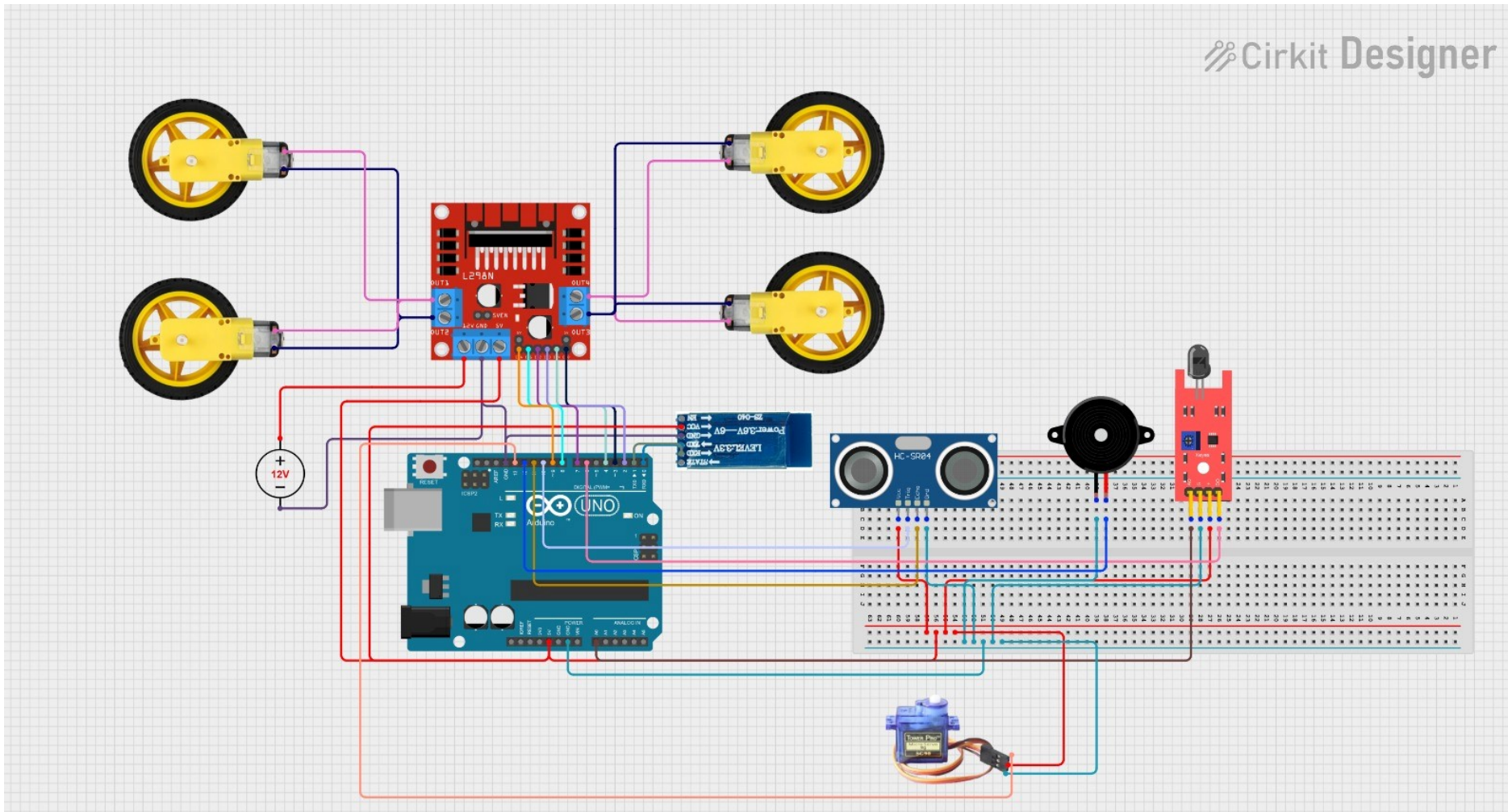


Fig 2: Interfacing Arduino Uno with L298N driver and Flame/Ultrasonic sensor array.



Fig 3: Final prototype

KEY REFERENCES

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